CS2201 Computer Organization & Architecture

L29: Non-Volatile Memory Technologies



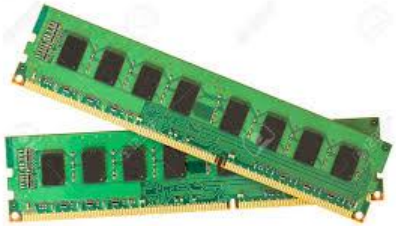
John Jose

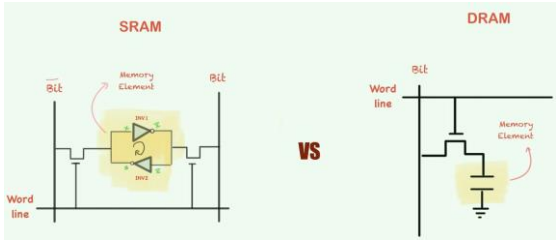
Professor

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Indian Institute of Technology Guwahati

RAM vs ROM

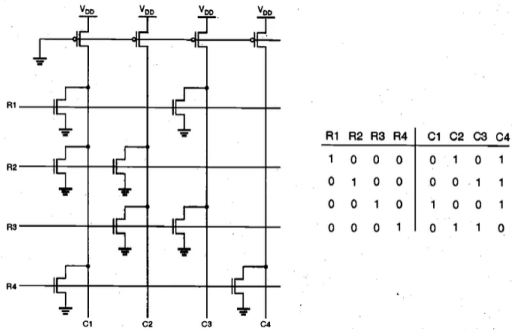




RAM (Random Access Memory)

- Volatile but faster

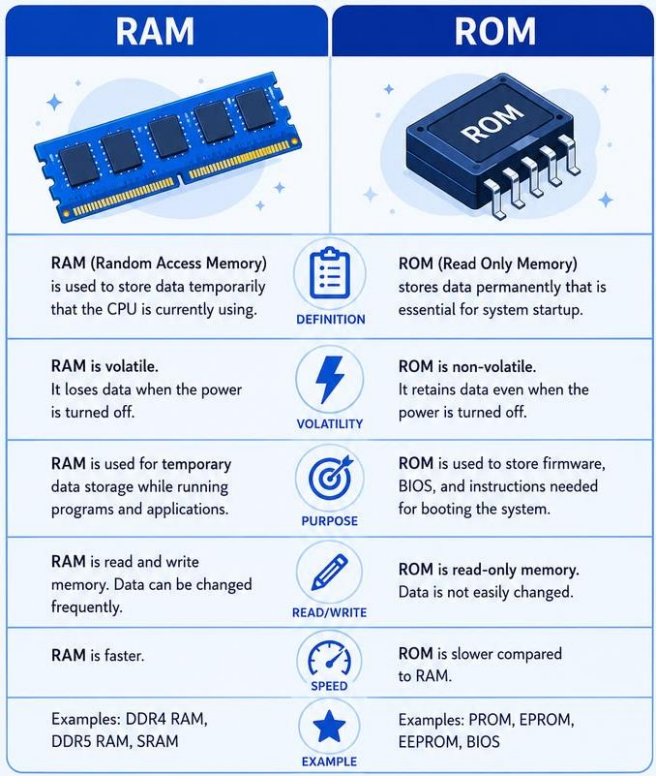




R1	R2	R3	R4	C1	C2	C3	C4
1	0	0	0	0	1	0	1
0	1	0	0	0	0	1	1
0	0	1	0	1	0	0	1
0	0	0	1	0	1	1	0

ROM (Read Only Memory)

- Non volatile but slower



Drawback of Memory Technologies

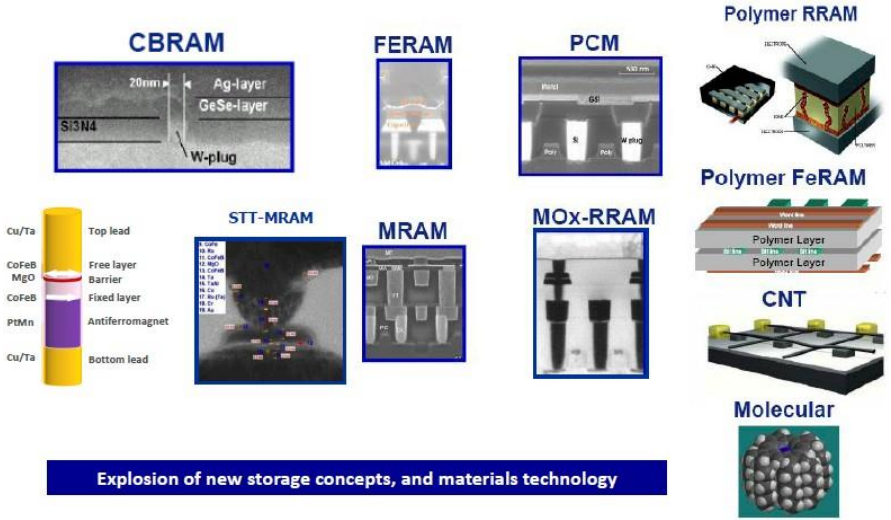
- Applications are becoming more data intensive
- Future high-end systems will be handling exabytes of data
- Use of conventional memory technologies will result in
  - low performance
  - high power consumption
- Solution - NVMs



Emerging Memory Technologies (NVM)

- Low power consumption
- High packaging density

- Low endurance
- High write latency

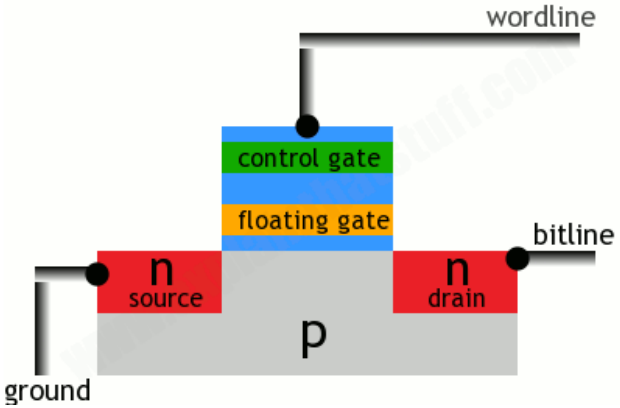


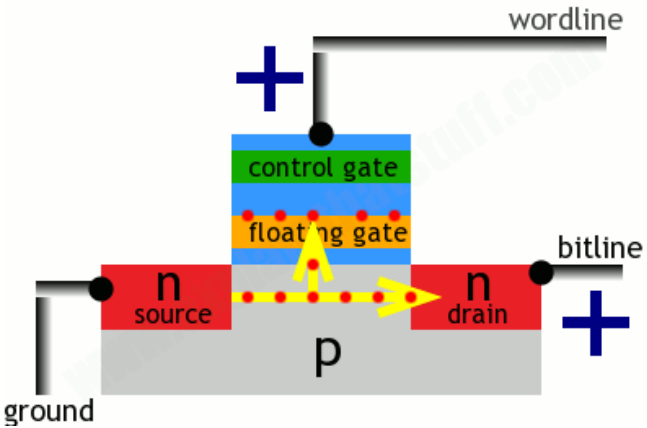
Types of Non Volatile Memories

- Flash Memory
- PCM (Phase Change Memory)
- STT- RAM (Spin Transfer Torque RAM)
- ReRAM (Resistive RAM)

Flash Memory

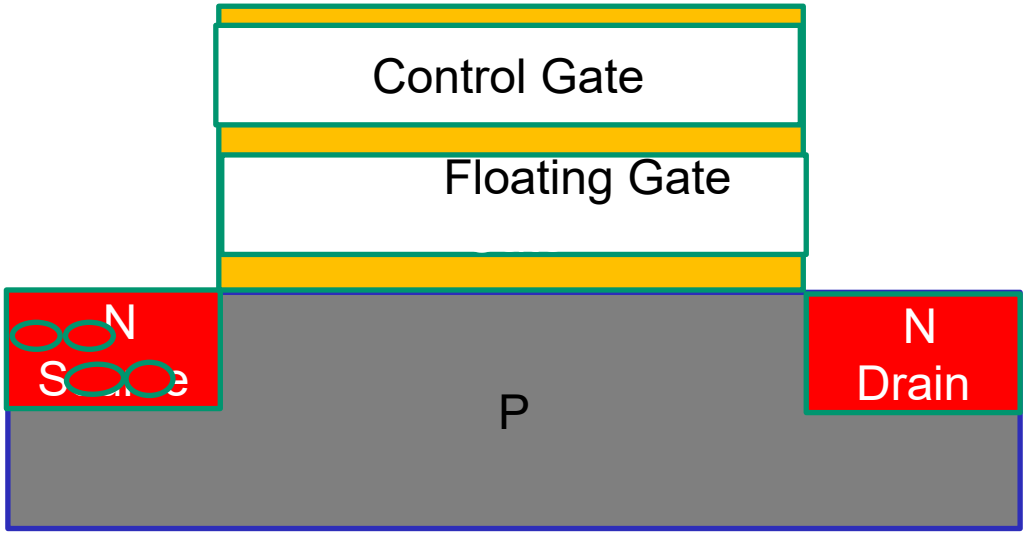
- Floating gate transistor





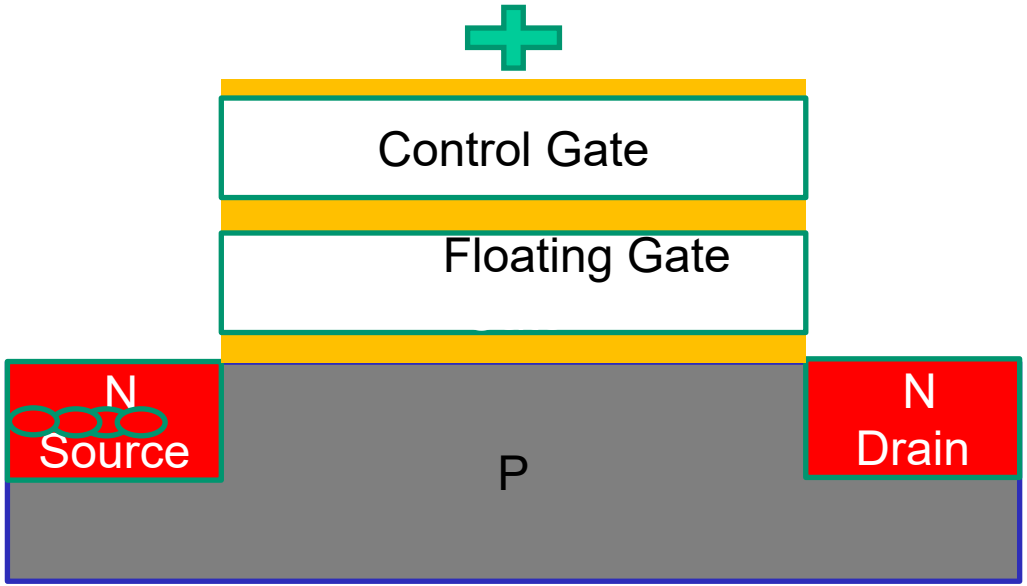
Flash Memory

Storing ‘1’

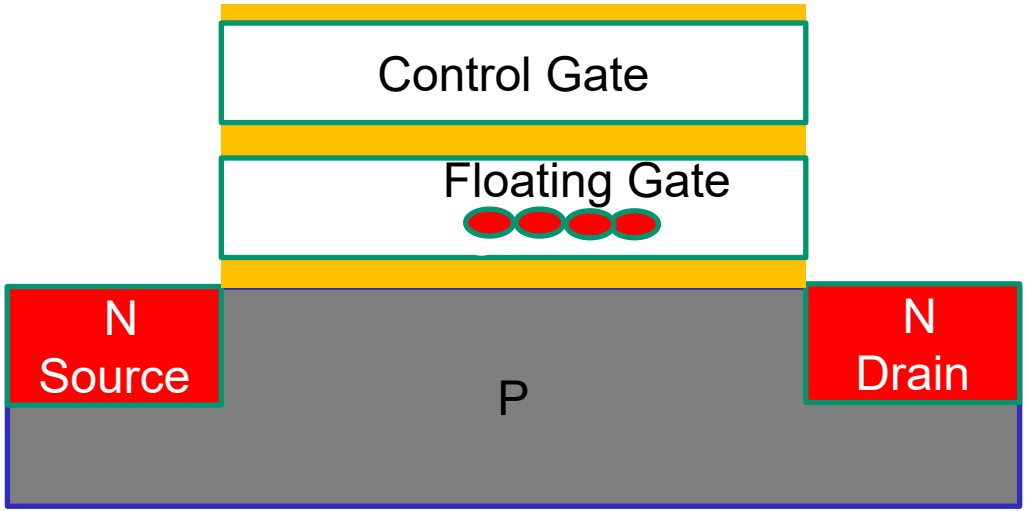


Flash Memory

Storing ‘0’

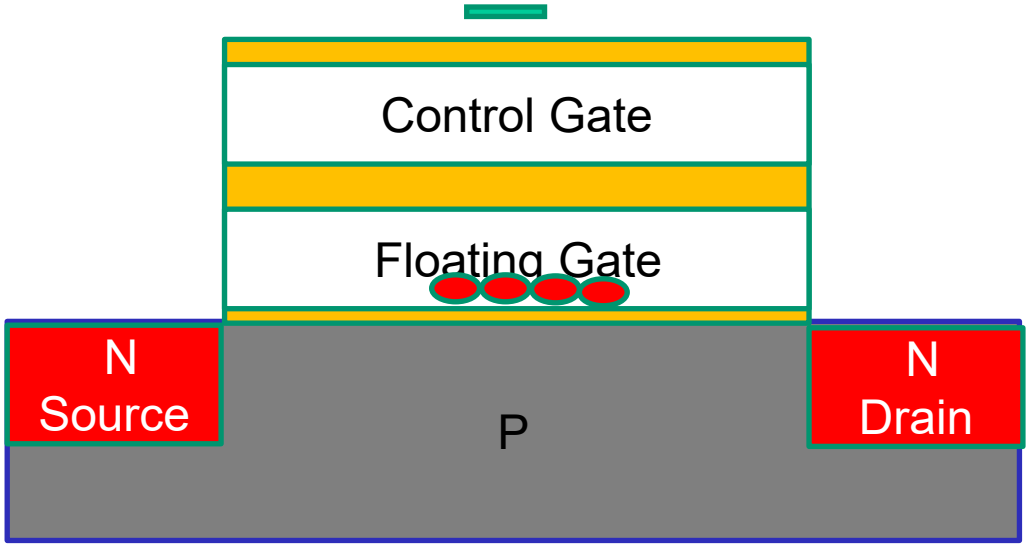


Storing ‘0’

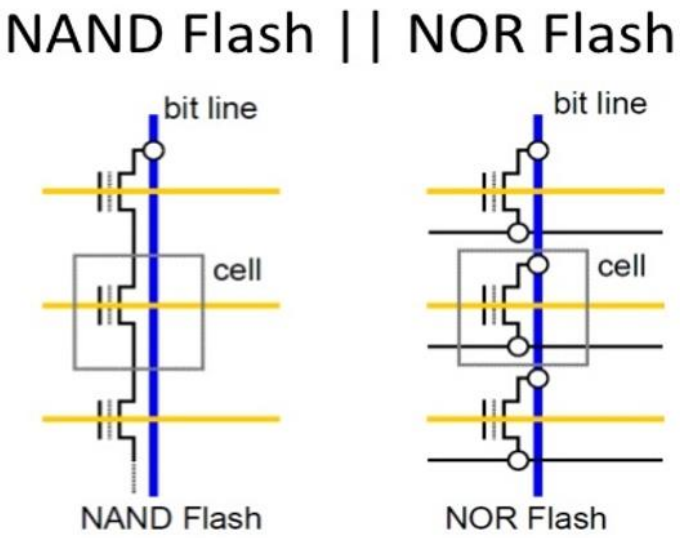


Flash Memory

Erasing

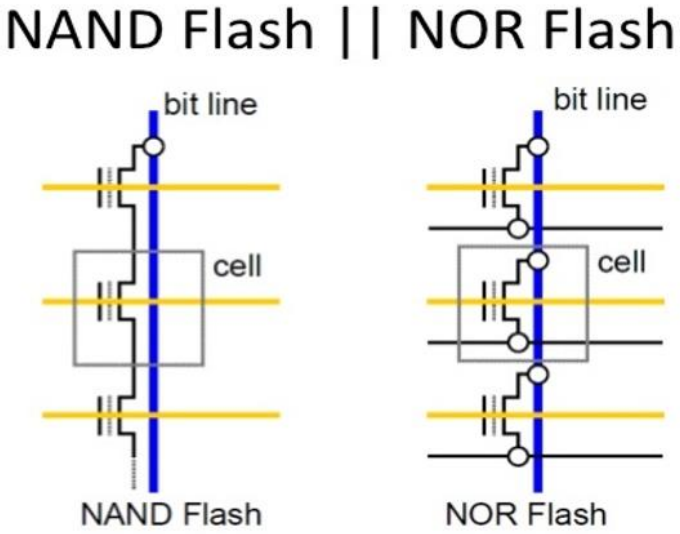


NAND vs NOR Flash Memory



- NAND and NOR flash are non-volatile memory technologies that retain data without power, differing primarily in how their internal transistors are wired and used.
- **NAND Flash:** Memory cells are linked in a **series (chain)**. This compact design uses fewer wires and smaller cell sizes, allowing for much higher storage
- **NOR Flash:** Memory cells are linked in parallel to the bit lines. This structure allows independent, direct access to every single byte or word.

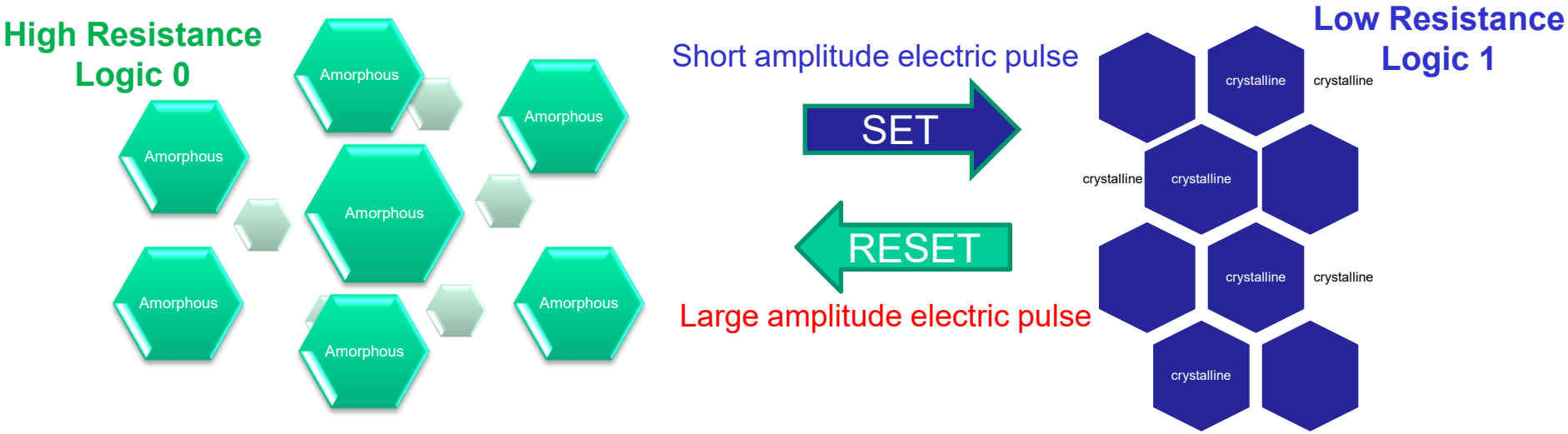
NAND vs NOR Flash Memory



Aspect	NAND Flash Memory	NOR Flash Memory
Architecture	Memory cells connected in **series (chain)**	Memory cells connected in **parallel (branch-like)**
Storage Density	**High** – more data can be stored in less space	**Low** – requires more space for same data
Cost per Bit	**Cheaper**, more cost-effective	**More expensive**
Read Speed	Slower random read; good for **sequential reads**	**Fast random read**, ideal for quick data access
Write Speed	Fast for writing large data blocks	Fast for writing small amounts of data
Erase Operation	Done at **block level** (large units)	Done at **byte or sector level** (fine-grained)
Endurance	Lower program/erase cycles	Higher endurance
Access Type	Sequential access	Random access
XIP (Execute-In-Place)	Not supported	Supports direct code execution
Best Used For	SSDs, memory cards, USB drives, smartphones	BIOS chips, microcontrollers, embedded firmware
Performance Focus	Maximum storage & low cost	Fast execution & reliability
Typical Capacity	High (GB–TB)	Low (KB–MB)

Phase Change Memory

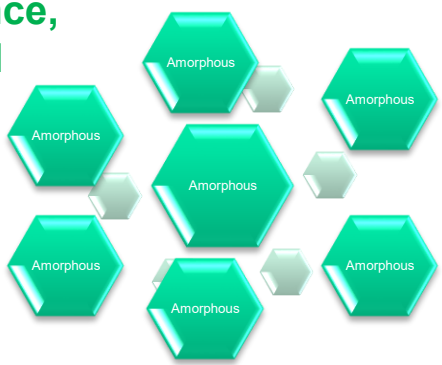
Phase-change memory (PCM) is a non-volatile random-access memory technology that stores digital data by changing the physical state of a material using heat



The data storage capability is achieved from the resistance differences between an amorphous (high resistance) and a crystalline (low-resistance) phase of the chalcogenide-based material.

Phase Change Memory

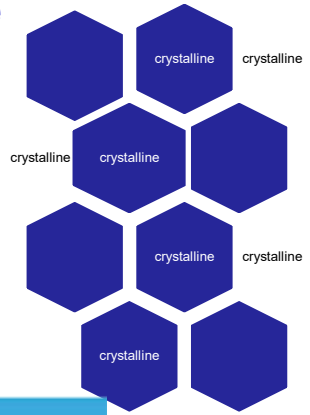
High Resistance, Disordered Logic 0

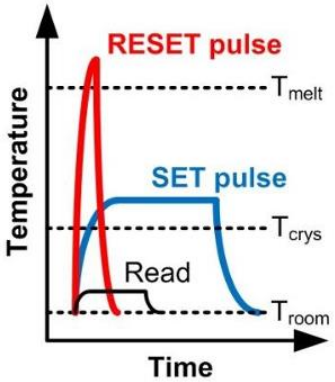


SET

RESET

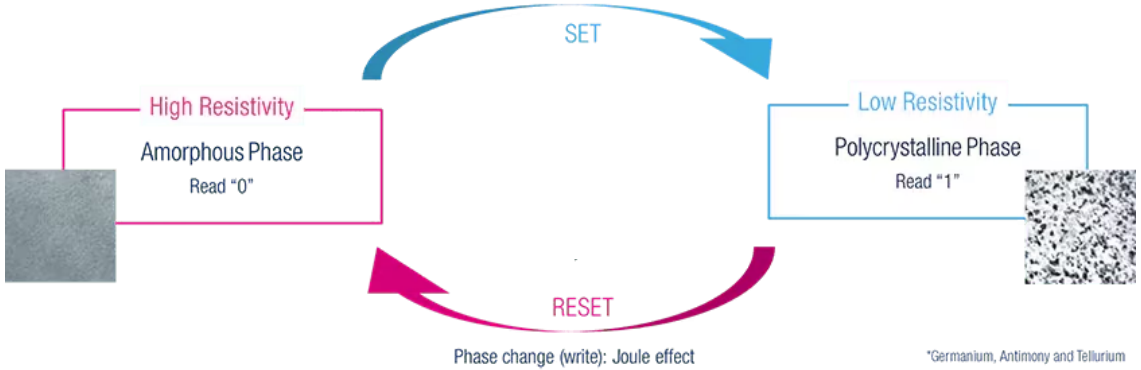
Low Resistance Ordered Logic 1





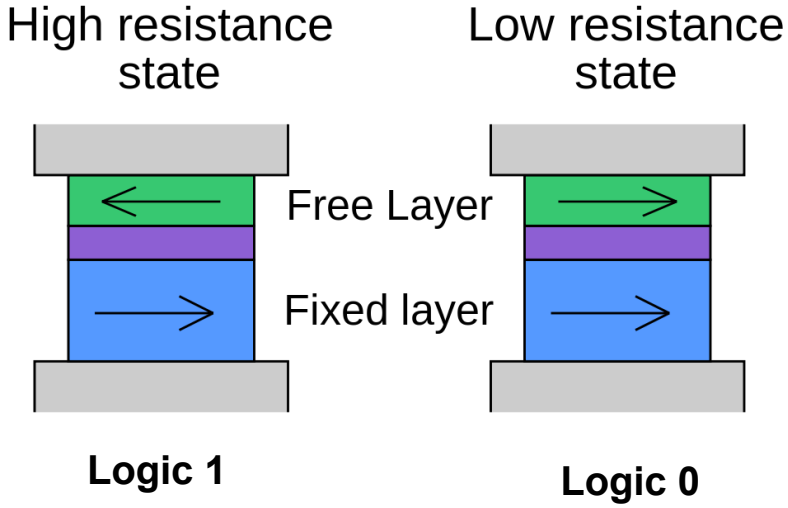
Storage Mechanism: Material with Variable Resistance

$Ge_2Sb_2Te_5$ (GST*) chalcogenide alloy has 2 phases with difference Resistivity



Spin Transfer Torque RAM

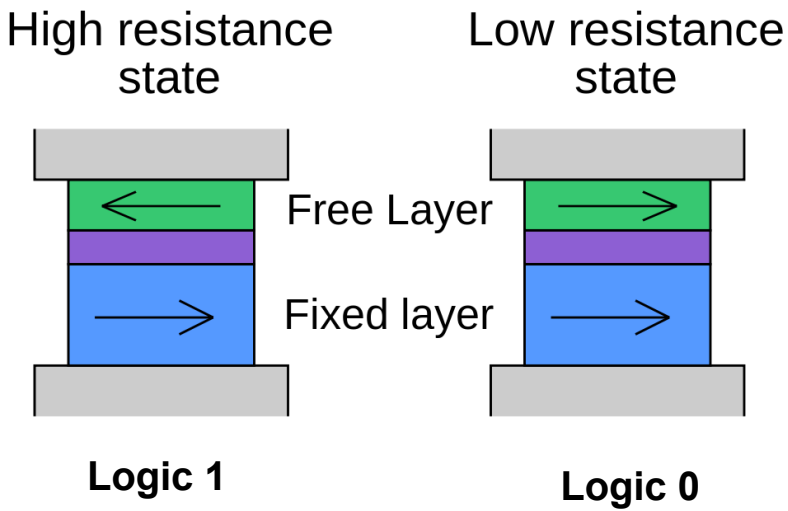
STT-MRAM (Spin-Transfer Torque Magnetic Random Access Memory) works by storing binary data (0 and 1) inside a magnetic storage element called a Magnetic Tunnel Junction (**MTJ**), using electron spin to switch magnetic orientations and tunneling resistance to read the state



STT-RAM cell - three-layer sandwich

Fixed (Reference) Layer: A ferromagnetic layer whose magnetic orientation is permanently pinned in one direction.
Tunnel Barrier: An extremely thin insulating layer (typically Magnesium Oxide, MgO).
Free Layer: A ferromagnetic layer whose magnetic orientation can be flipped.

Spin Transfer Torque RAM



STT-RAM cell - three-layer sandwich

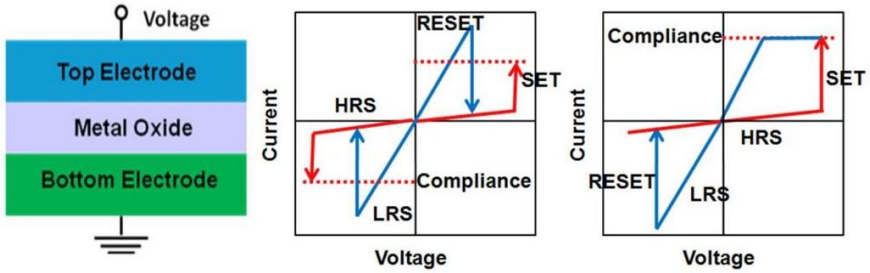
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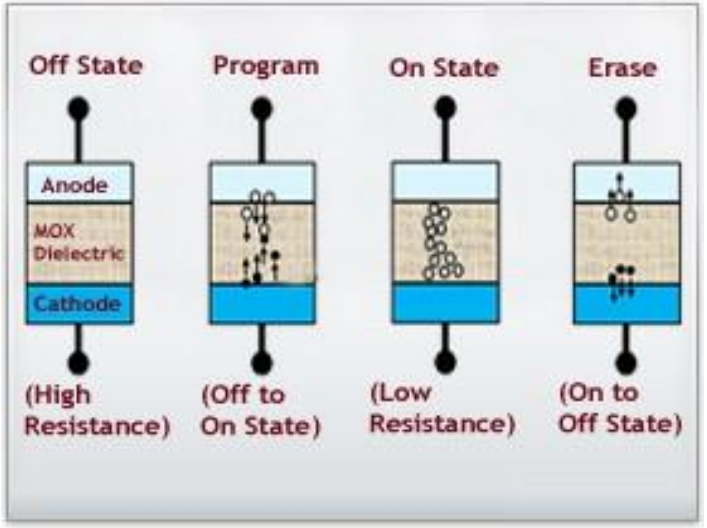
Resistive RAM (ReRAM)

- ReRAM memory cell is a metal-insulator-metal (MIM) structure with a metal-oxide layer sandwiched between top metal electrode (TE) and bottom metal electrode (BE).
- They can switch between high resistance state (HRS) and low resistance state (LRS) by applying current or voltage pulses to store 0 and 1 bits.
- The switching from HRS to LRS is called the “set” process and from LRS to HRS is called the “reset” process



Resistive RAM (ReRAM)

- High resistance : 0
- Low resistance : 1
- OFF to ON : 1
- ON to OFF : 0



Comparison Between NVMs

	SRAM	DRAM	Flash	PCM	STT-RAM	ReRAM
Cell size (F^2) *F denotes feature size	120-200	6-10	4-6	4-12	6-50	4-10
Access granularity	64 B	64 B	4 KB	64 B	64 B	64 B
Read latency	1-10 ns	50 ns	25 µs	50 ns	10 ns	10 ns
Write latency	1-10 ns	50 ns	500 µs	500 ns	50 ns	50 ns
Erase latency	N/A	N/A	2 ms	N/A	N/A	N/A
Endurance	$> 10^{15}$	$> 10^{15}$	$10^4 - 10^5$	$10^8 - 10^9$	$> 10^{12}$	10^{11}
Standby power	60 nW	Refresh power	0	0	0	0

Characteristics of NVMs

- High packing density
- Low power consumption
- Data retention
- Limited write endurance
- High write latency
- High energy for writes



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